

Operating System

slides contains content adapted from John Kubiatawicz (Berkeley), John Ousterhout(Stanford)

Introduction

- What is an Operating System?
- Mainframe Systems
- Desktop Systems
- Multiprocessor Systems
- Distributed Systems
- Clustered System
- Real -Time Systems
- Handheld Systems
- Computing Environments

What is an Operating System?

- A program that acts as an intermediary between a user of a computer and the computer hardware.

Operating System Definitions

- **Resource allocator** – manages and allocates resources.
- **Control program** – controls the execution of user programs and operations of I/O devices .
- **Kernel** – the one program running at all times.

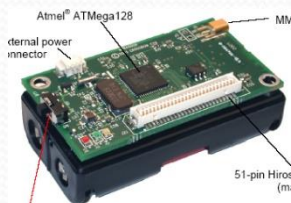
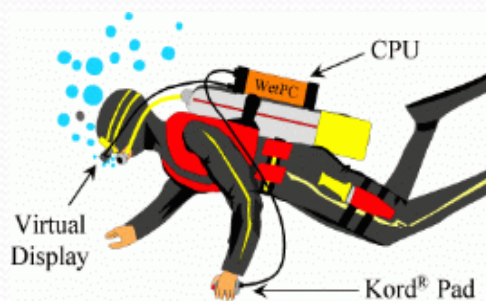
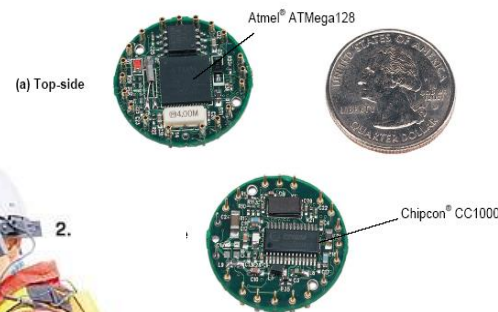
Goals of an Operating System

- Simplify the execution of user programs and make solving user problems easier.
- Use computer hardware efficiently.
 - Allow sharing of hardware and software resources.
- Make application software portable and versatile.
- Provide isolation, security and protection among user programs.
- Improve overall system reliability
 - error confinement, fault tolerance, reconfiguration.

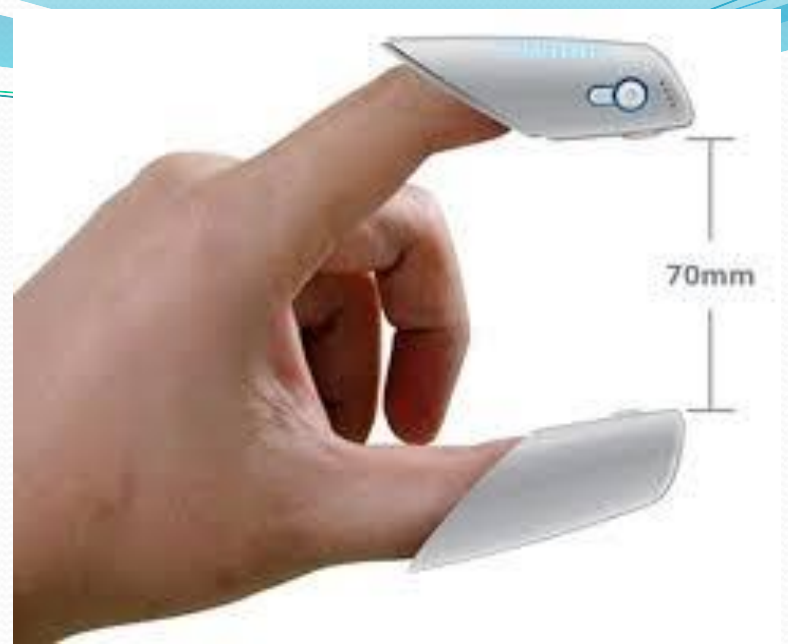
Why should I study Operating Systems?

- Need to understand interaction between the hardware and applications
 - New applications, new hardware..
 - Inherent aspect of society today
- Need to understand basic principles in the design of computer systems
 - efficient resource management, security, flexibility
- Increasing need for specialized operating systems
 - e.g. embedded operating systems for devices - cell phones, sensors and controllers
 - real-time operating systems – vehicles, aircraft control, multimedia services

Systems Today and The Future



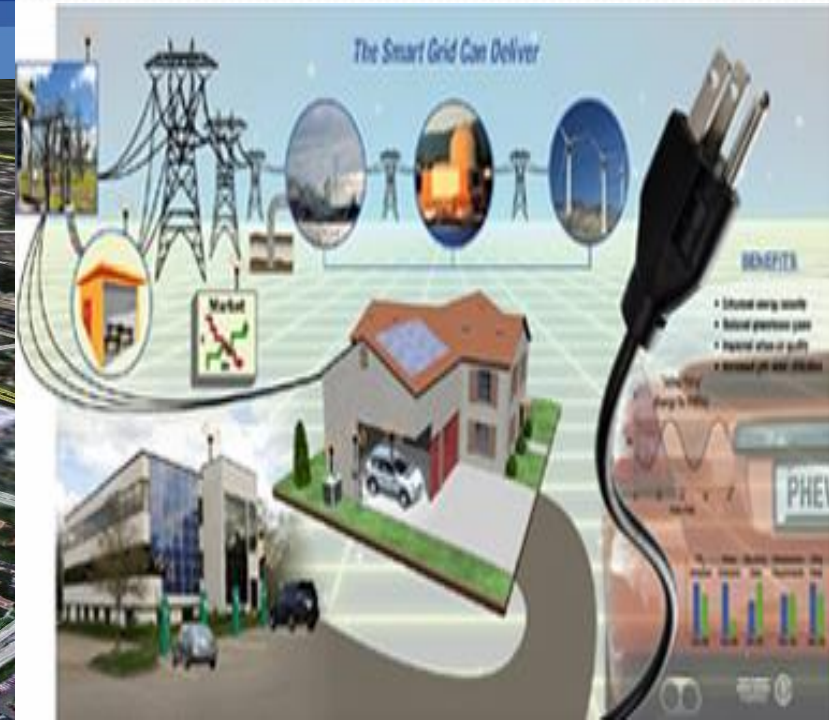
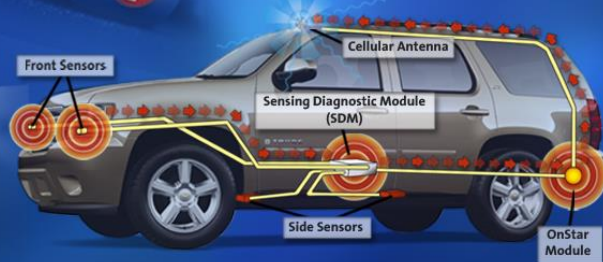




General Motors

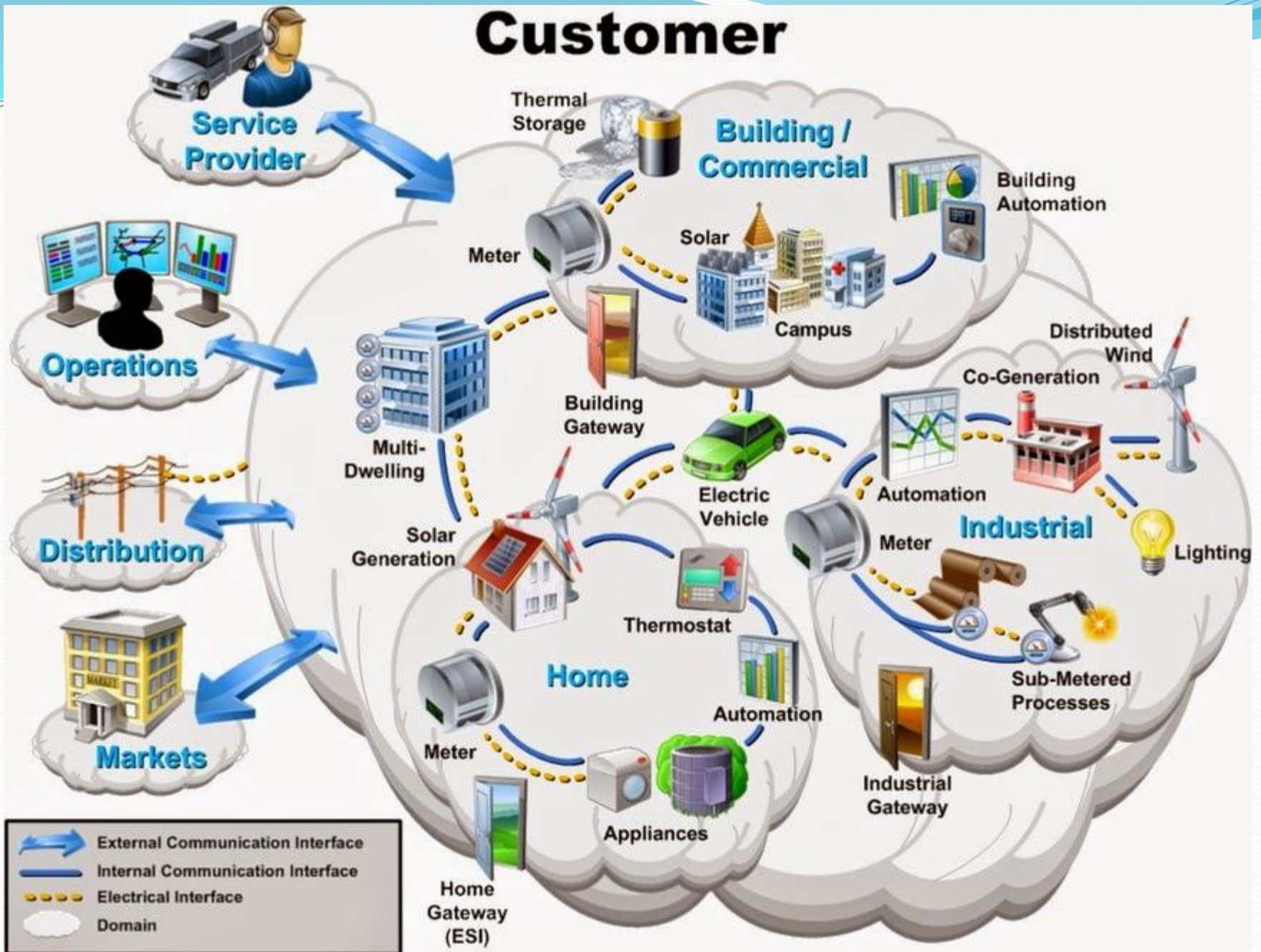
OnStar

by GM



Irvine Sensorium

Customer



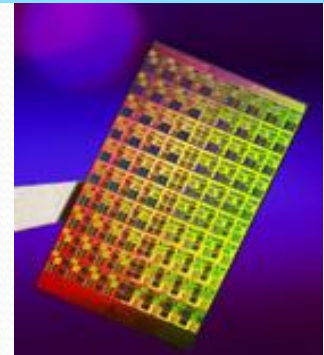
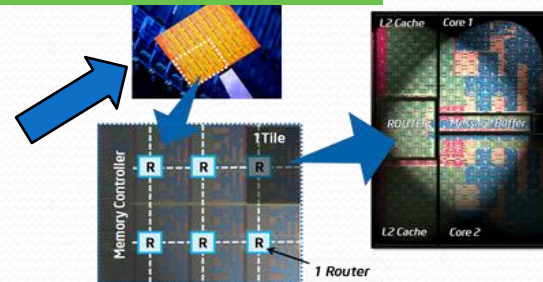
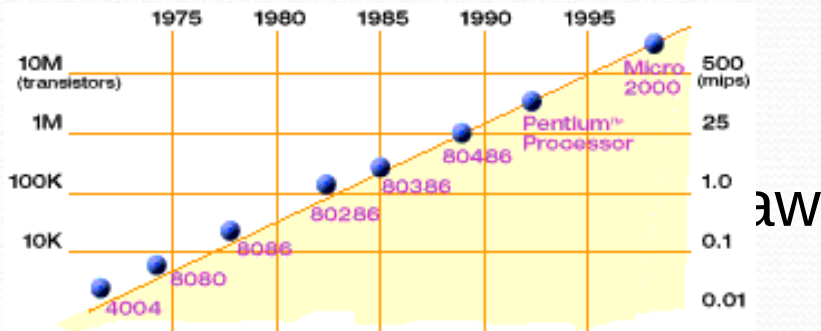
Hardware Complexity Increases

Moore's Law: 2X

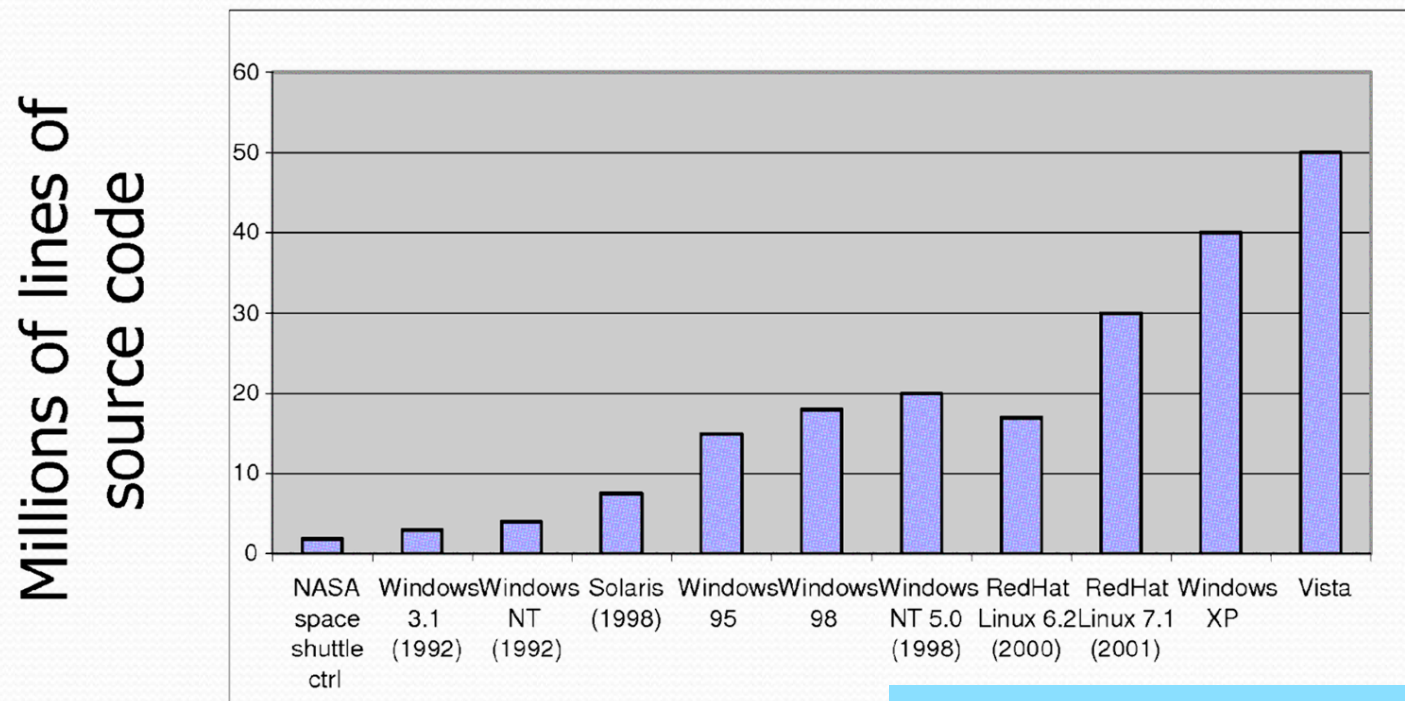
transistors/Chip Every 1.5 years

From Berkeley OS course

Intel Multicore Chipsets



Software Complexity Increases

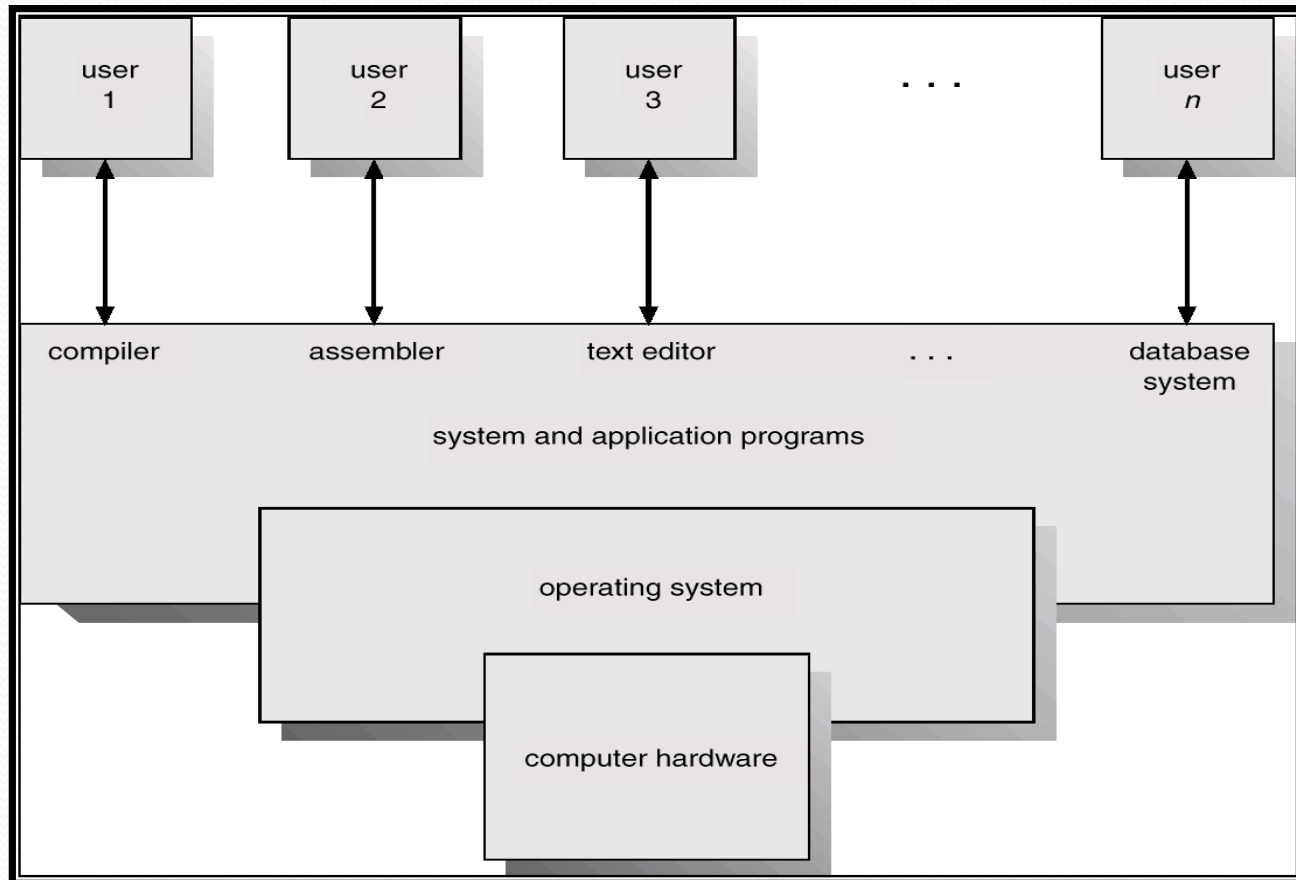


From MIT's 6.033 course

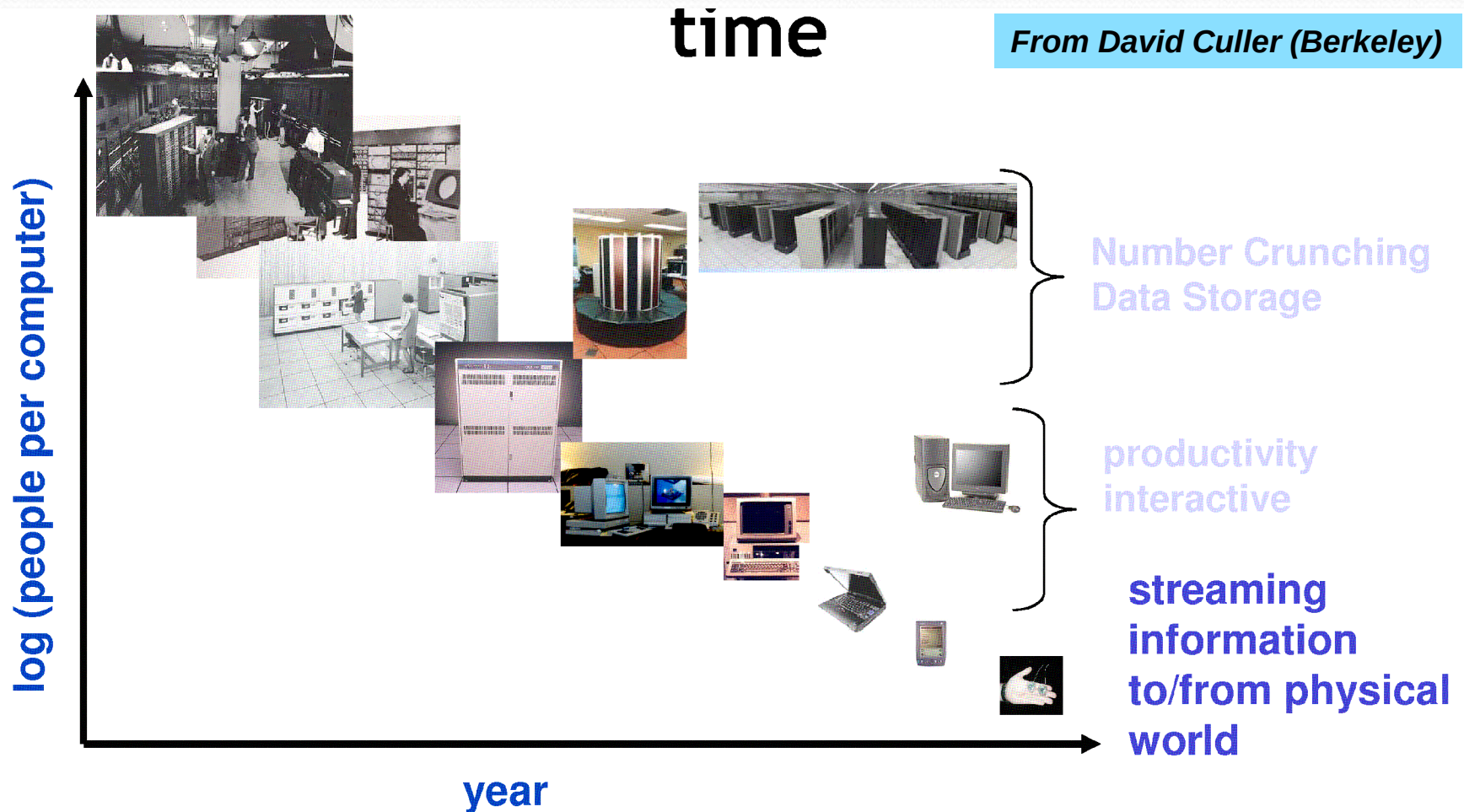
Computer System Components

1. Hardware – provides basic computing resources (CPU, memory, I/O devices).
2. Operating system – controls and coordinates the use of the hardware among the various application programs for the various users.
3. Applications programs – define the ways in which the system resources are used to solve the computing problems of the users (compilers, database systems, video games, business programs).
4. Users (people, machines, other computers).

Abstract View of System Components



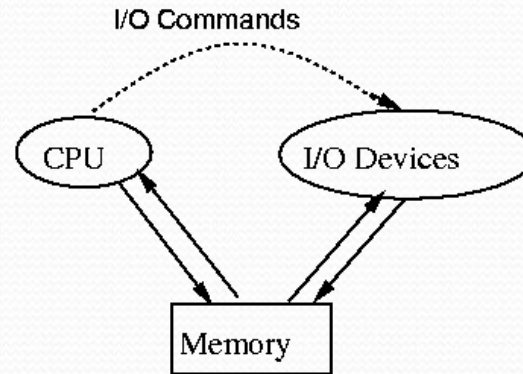
People-to-Computer Ratio Over Time



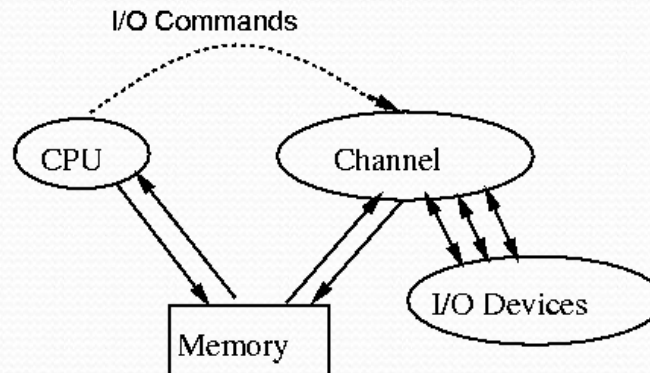
Speeding up I/O-

DMA is a method of transferring data from the computer's RAM to another part of the computer without processing it using the CPU

- Direct Memory Access (DMA)



- Channels



Batch Systems - I/O completion

- How do we know that I/O is complete?
 - Polling or busy waiting:
 - Device sets a flag when it is busy.
 - Program tests the flag in a loop waiting for completion of I/O.
 - Interrupts:
 - On completion of I/O, device forces CPU to jump to a specific instruction address that contains the interrupt service routine.
 - After the interrupt has been processed, CPU returns to code it was executing prior to servicing the interrupt.

Multiprogramming and Timesharing

- **Multiprogramming**

- Single user cannot keep CPU and I/O devices busy at all times
- Multiprogramming organizes jobs (code and data) so CPU always has -one to execute
- A subset of total jobs in system is kept in memory
- One job selected and run via **job scheduling**
- When it has to wait (for I/O for example), OS switches to another job

• **Timesharing (multitasking)** is logical extension of multiprogramming in which CPU switches jobs so frequently that users can interact with each job while it is running, creating interactive-computing

- Response time should be < 1 second**
 - Each user has at least one program executing in memory → **process**
 - If several jobs ready to run at the same time → **CPU scheduling**
 - If processes don't fit in memory, **swapping** moves them in and out to run
- **Virtual memory** allows execution of processes not completely in memory

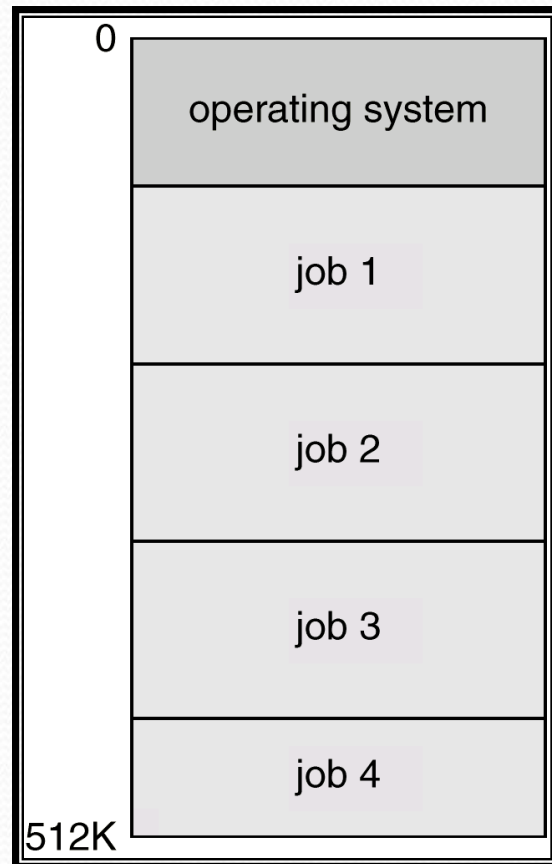
multiprogramming VS time sharing

Main difference between multiprogramming and time sharing is that

- **multiprogramming** is the effective utilization of CPU time, by allowing several programs to use the CPU at the same time but
- **time sharing** is the sharing of a computing facility by several users that want to use the same facility at the same time. Each user on a time sharing system gets her own terminal and gets the feeling that she is using the CPU alone

Multiprogrammed Batch Systems

Several jobs are kept in main memory at the same time, and the CPU is multiplexed among them.



OS Features Needed for Multiprogramming

- I/O routine supplied by the system.
- Memory management – the system must allocate the memory to several jobs.
- CPU scheduling – the system must choose among several jobs ready to run.
- Allocation of devices.

Desktop Systems

- *Personal computers* – computer system dedicated to a single user.
- I/O devices – keyboards, mice, display screens, small printers.
- User convenience and responsiveness.
- Can adopt technology developed for larger operating system' often individuals have sole use of computer and do not need advanced CPU utilization of protection features.
- May run several different types of operating systems (Windows, MacOS, UNIX, Linux)

Parallel Systems

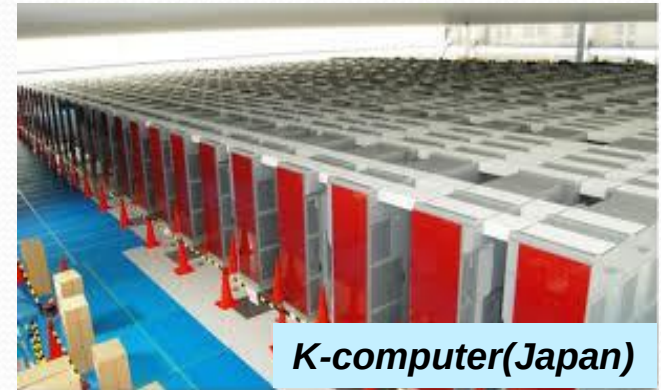
- Multiprocessor systems with more than one CPU in close communication.
- ***Tightly coupled system*** – processors share memory and a clock; communication usually takes place through the shared memory.
- Advantages of parallel system:
 - Increased *throughput*
 - Economical
 - Increased reliability

Parallel Computing Systems

ILLIAC 2 (Uillinois)



*Climate modeling,
earthquake
simulations,
genome analysis,
protein folding,
nuclear fusion
research,*



K-computer(Japan)



**Connection Machine
(MIT)**

Tianhe-1(China)

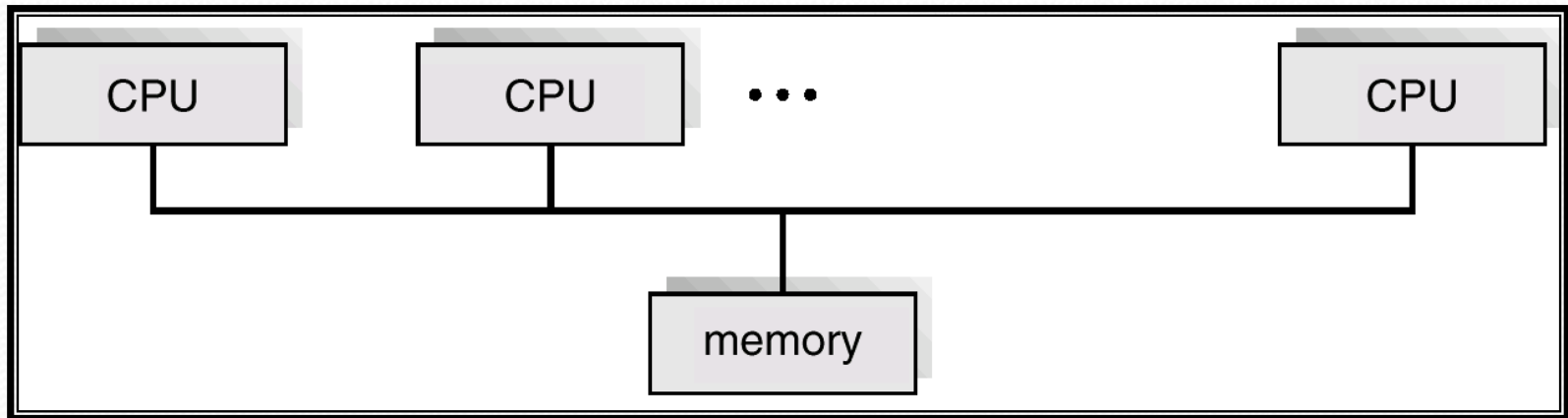


IBM Blue Gene

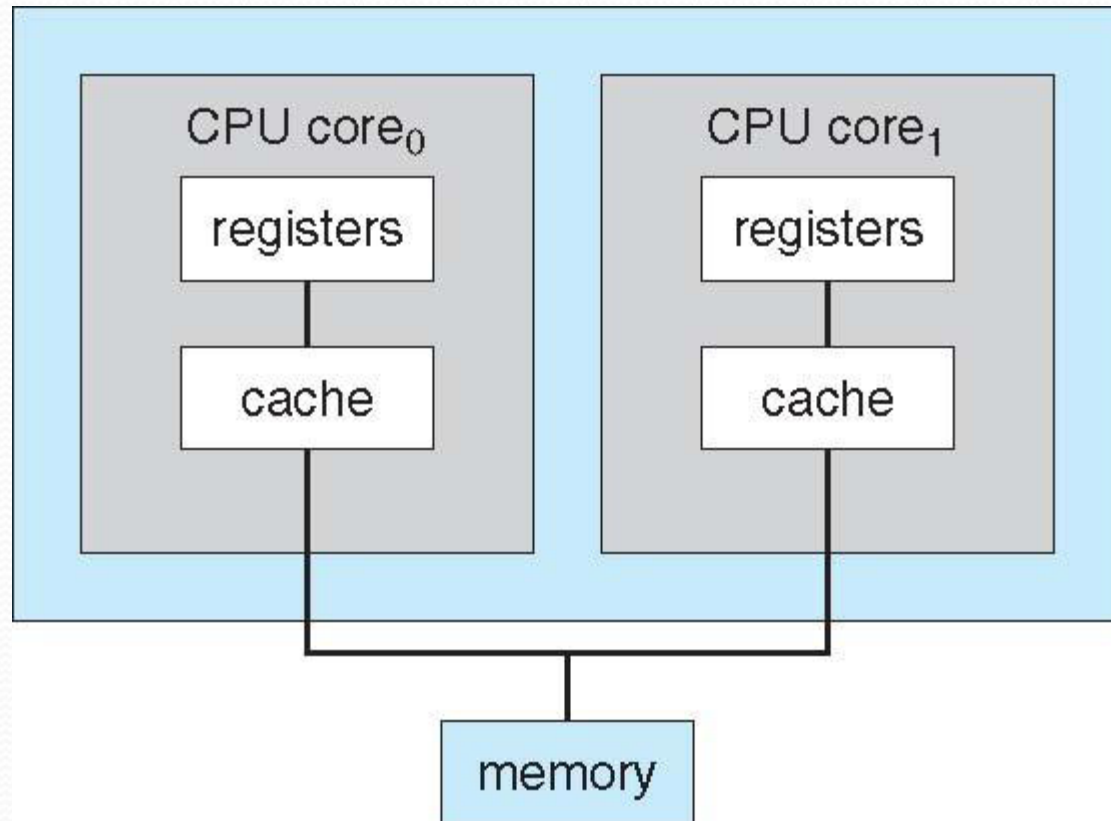
Parallel Systems (Cont.)

- *Symmetric multiprocessing (SMP)*
 - Each processor runs an identical copy of the operating system.
 - Many processes can run at once without performance deterioration.
 - Most modern operating systems support SMP
- *Asymmetric multiprocessing*
 - Each processor is assigned a specific task; master processor schedules and allocated work to slave processors.
 - More common in extremely large systems

Symmetric Multiprocessing Architecture

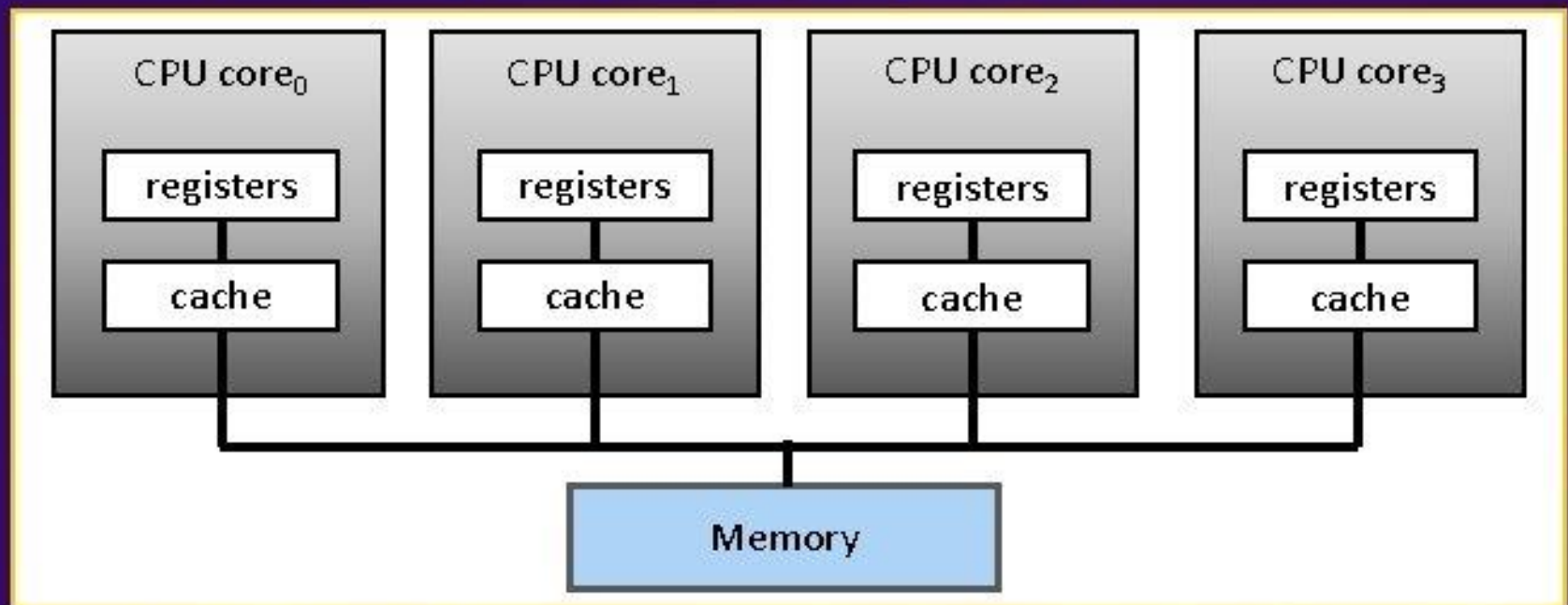


Dual Core Design



Quad-Core Design

- ◆ An MPU chip has **four** CPU cores, each:
 - ✓ owning its own small L1 cache,
 - ✓ sharing a large L2 cache, and
 - ✓ accessing the main memory through L2.



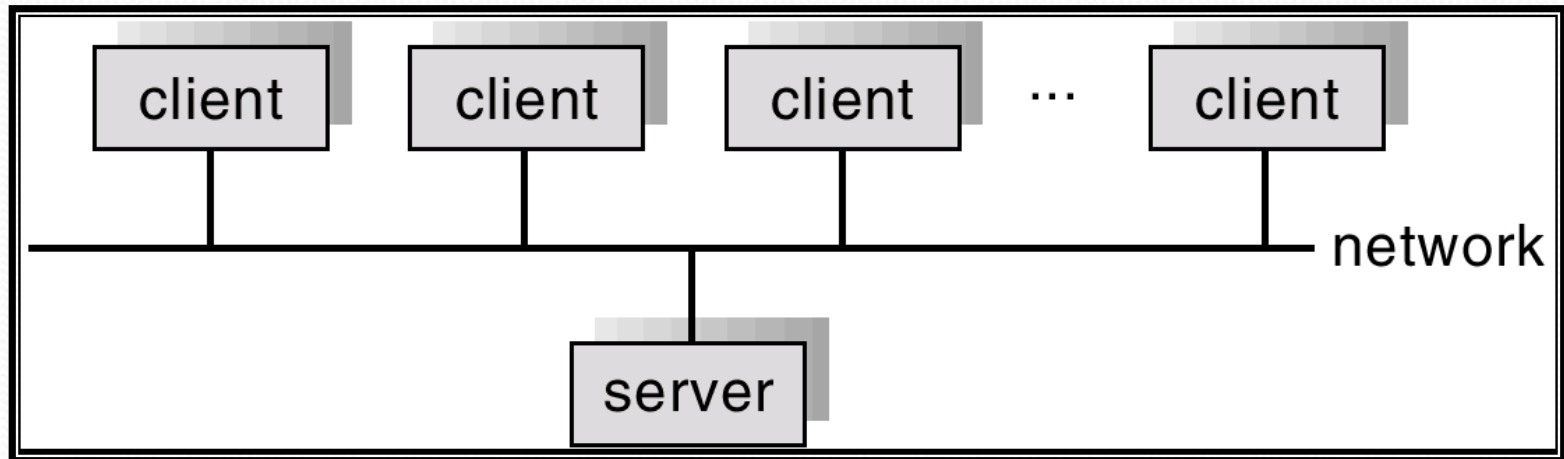
Distributed Systems

- Distribute the computation among several physical processors.
- *Loosely coupled system* – each processor has its own local memory; processors communicate with one another through various communications lines, such as high-speed buses or telephone lines.
- Advantages of distributed systems.
 - Resources Sharing
 - Computation speed up – load sharing
 - Reliability

Distributed Systems (cont)

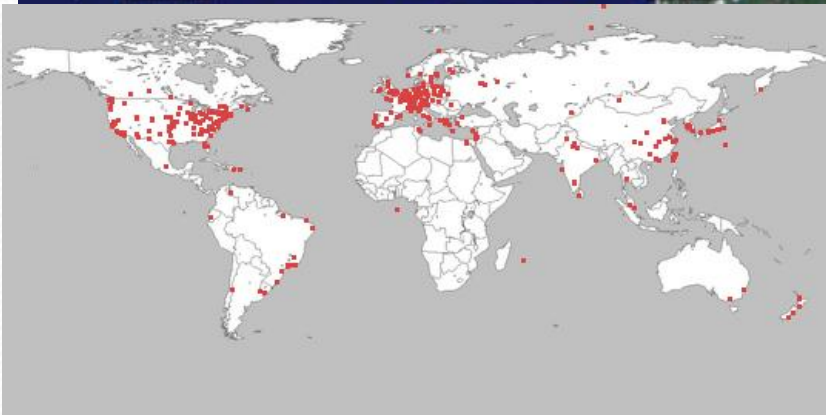
- Requires networking infrastructure.
- Local area networks (LAN) or Wide area networks (WAN)
- May be either client-server or peer-to-peer systems.

General Structure of Client-Server



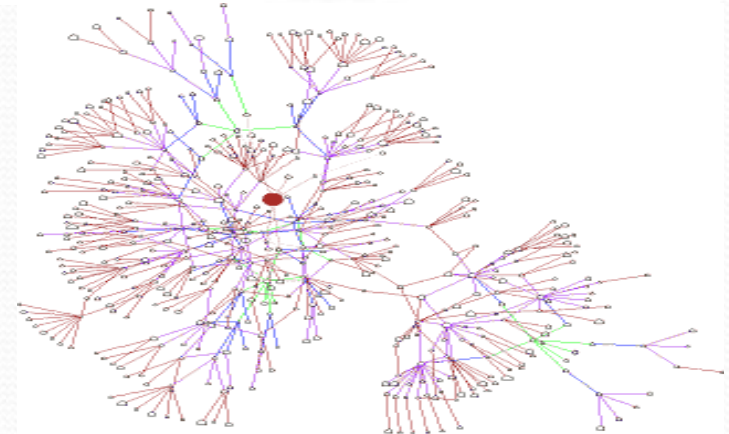
Distributed Computing Systems

Globus Grid Computing Toolkit



PlanetLab

Cloud Computing Offerings



Gnutella P2P Network

Serverless VS cloud computing

- Serverless is a cloud-native development model that allows developers to build and run applications without having to manage servers
- Serverless differs from other cloud computing models in that the cloud provider is responsible for managing both the cloud infrastructure and the scaling of apps. Serverless apps are deployed in containers that automatically launch on demand when called.
- Under a serverless model, a cloud provider runs physical servers and dynamically allocates their resources on behalf of users who can deploy code straight into production.

Cloud Computing Services: Who Manages What?

	Traditional IT	IaaS	PaaS	Serverless	SaaS
Applications					
Data					
Runtime					
Middleware					
OS					
Virtualization					
Servers					
Storage					
Networking					

 You manage  Provider manages

Societal Scale Information Systems

- The world is a large distributed system
 - Microprocessors in everything
 - Vast infrastructure behind them

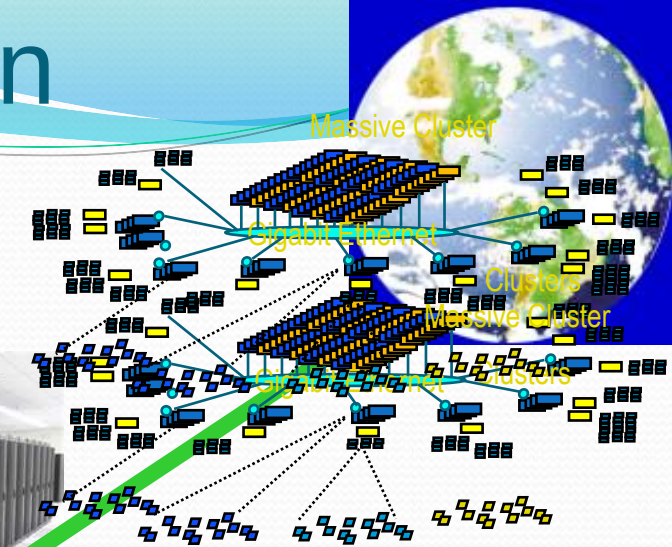
Internet
Connectivity



Micro-electromechanical
systems (MEMS) for
Sensor Nets

Scalable, Reliable,
Secure Services

Databases
Information Collection
Remote Storage
Online Games
Commerce
...



Clustered Systems

- Clustering allows two or more systems to share storage.
- Provides high reliability.
- *Asymmetric clustering*: one server runs the application while other servers standby.
- *Symmetric clustering*: all N hosts are running the application.

Real-Time Systems

- Often used as a control device in a dedicated application such as controlling scientific experiments, medical imaging systems, industrial control systems, and some display systems.
- Well-defined fixed-time constraints.
- Real-Time systems may be either *hard* or *soft* real-time.

Clustered System VS Multiprocessor System?

- - In Multiprocessor System Multiprocessors are tied together and in clustered system individual systems are tied together.
- - LAN networking is used in Clustered system, in Multiprocessing system LAN etc is not required.
- - The aim is high availability in clustered system whereas in multiprocessing it is time saving.

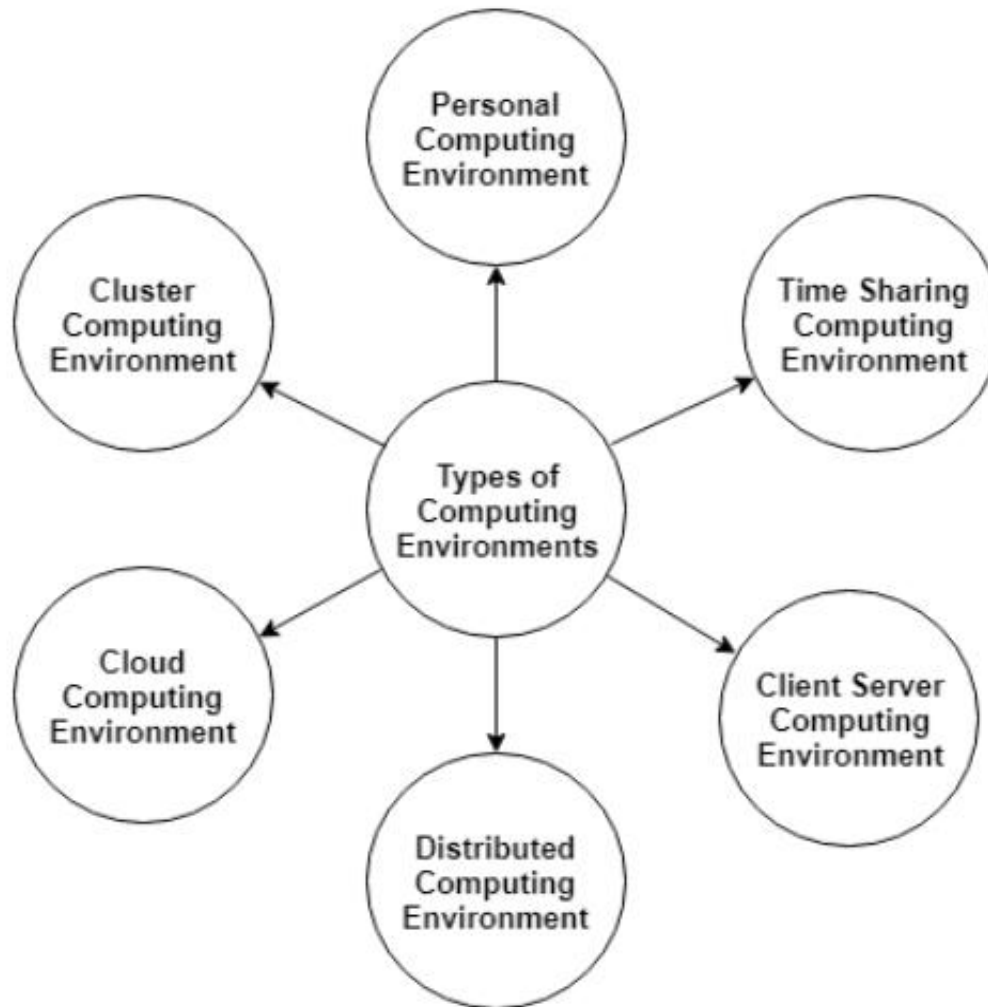
Real-Time Systems (Cont.)

- Hard real-time:
 - Secondary storage limited or absent, data stored in short term memory, or read-only memory (ROM)
- Soft real-time
 - Limited utility in industrial control of robotics
 - Useful in applications (multimedia, virtual reality) requiring advanced operating-system features.

Handheld Systems

- Personal Digital Assistants (PDAs)
- Cellular telephones
- Issues:
 - Limited memory
 - Slow processors
 - Small display screens.

Computing Environments



Computing Environments

- Traditional computing
- Client Server Computing
- Peer to Peer Computing
- Web-Based Computing

Traditional

- Traditional computing environment are blurring
- Office environment
 - PCs connected to a network, terminals attached to mainframe or minicomputers providing batch and timesharing
 - portals allowing networked and remote systems access to same resources
- Home networks
 - Used to be single system, then modems
 - Now firewalled, networked

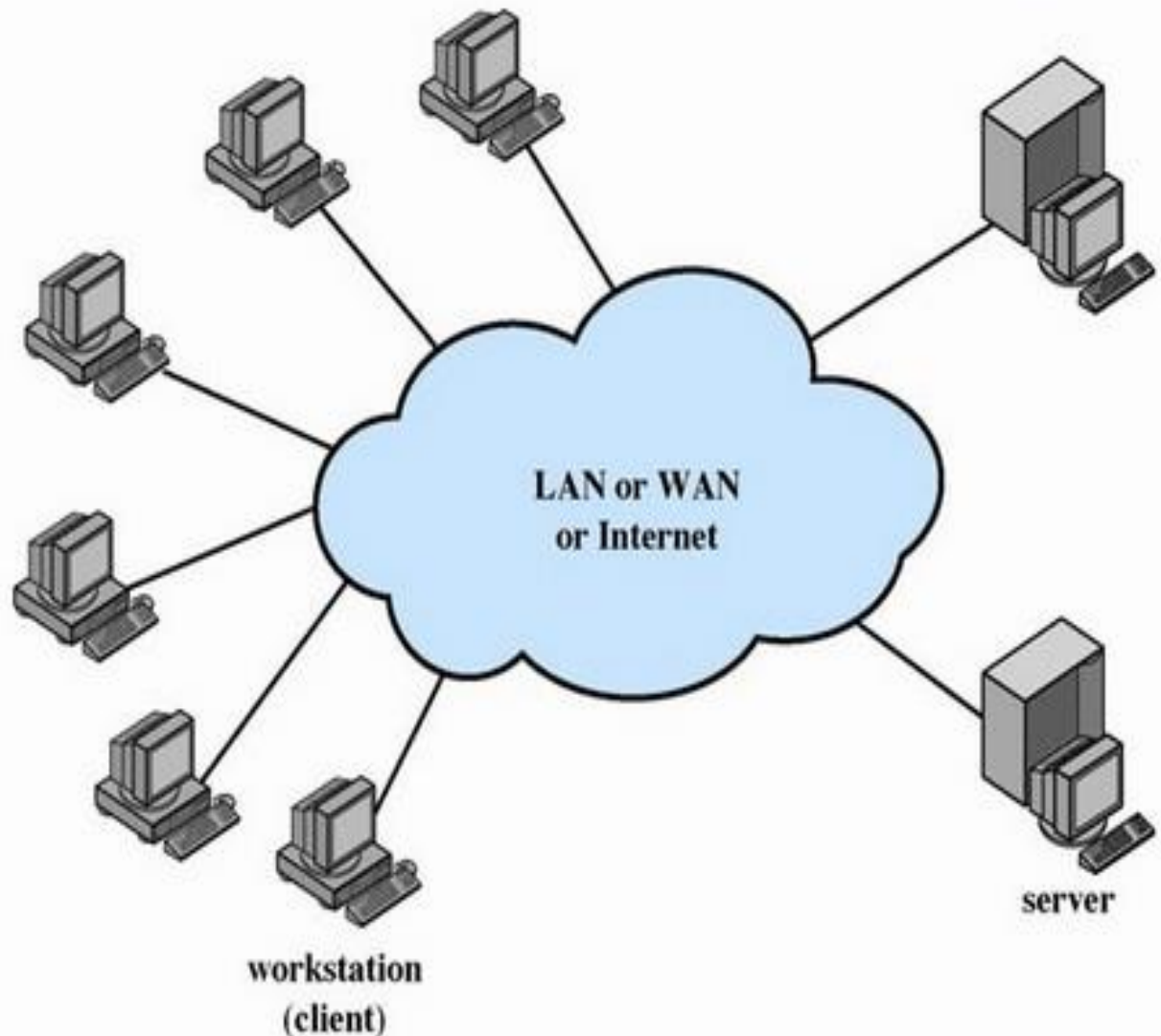
Client Server Computing

- Client-Server Computing
 - Dumb terminals supplanted by smart PCs
 - Many systems now **servers**, responding to requests generated by **clients**
 - **Compute-server** provides an interface to client to request services (i.e. database)
 - **File-server** provides interface for clients to store and retrieve files

Client/Server Computing



- **Client machines:**
single-user PCs or workstations that provide a highly *user-friendly interface* to the end user
- Each **server** provides a set of shared user services to the clients
- The server enables **many clients** to share access to the same database and enables the use of a high-performance computer system to manage the database

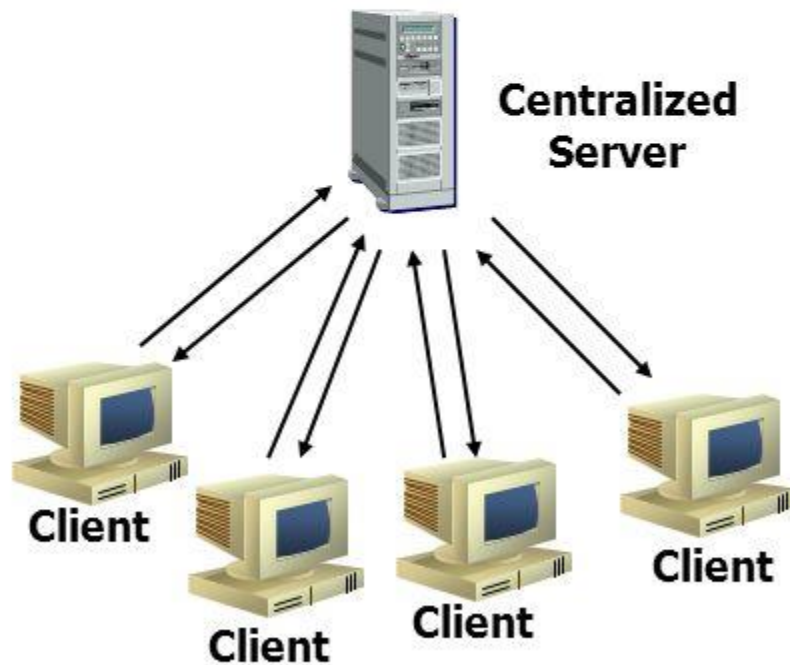


Peer to Peer Computing

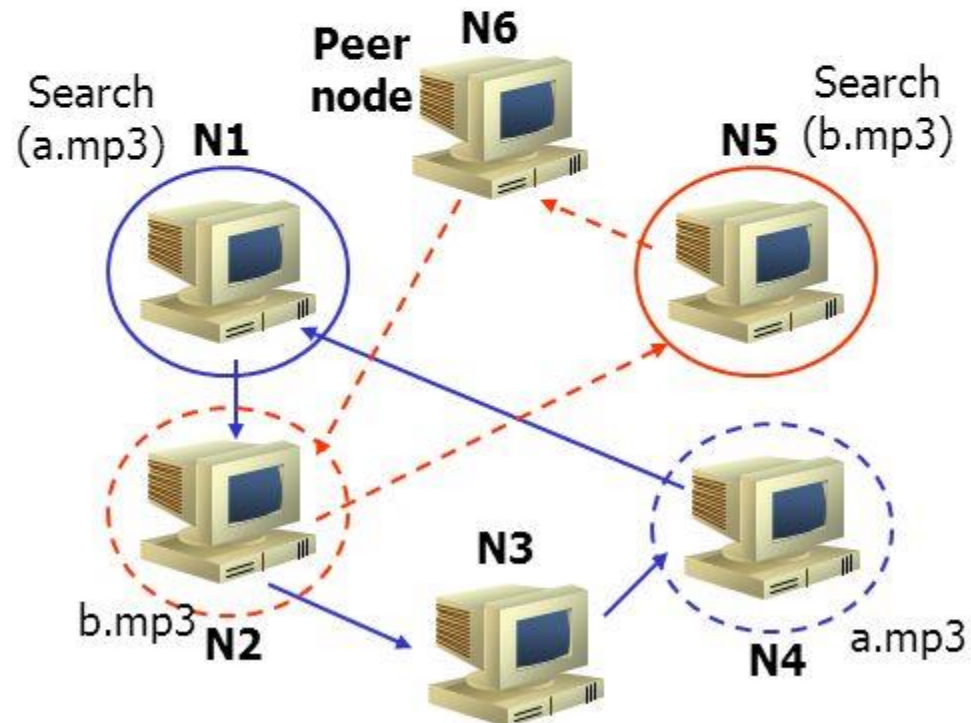
- Another model of distributed system
- P2P does not distinguish clients and servers
 - Instead all nodes are considered peers
 - May each act as client, server or both
 - Node must join P2P network
 - Registers its service with central lookup service on network, or
 - Broadcast request for service and respond to requests for service via **discovery protocol**

What is P2P Computing?

- What is P2P computing?
 - New computing paradigm
 - Every node is both client and server
 - Sharing computing resources(e.g., storage, CPU power etc) among peer nodes



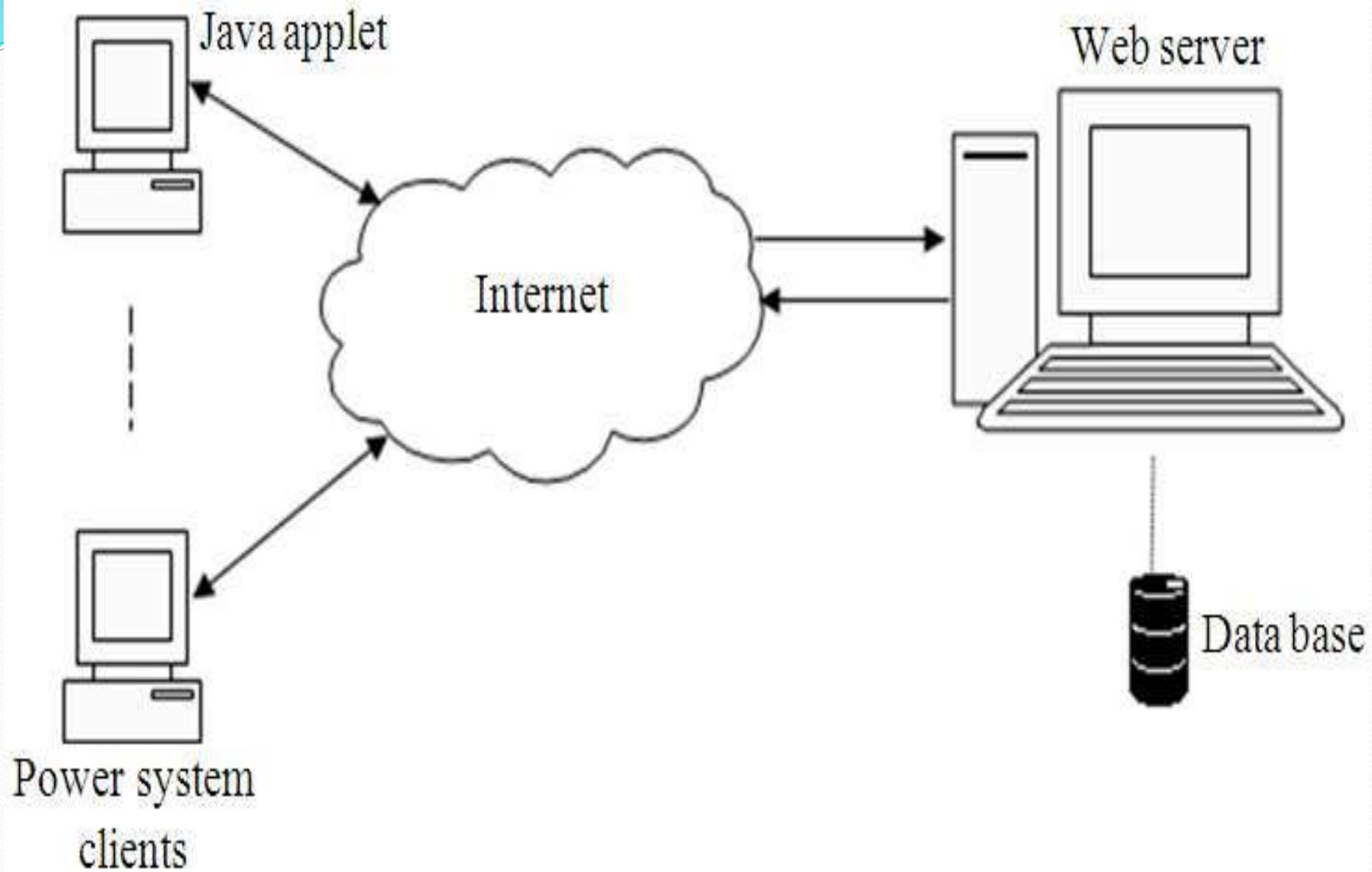
Client/Server Computing



Peer-to-Peer Computing

Web-Based Computing

- Web has become ubiquitous
- PCs most common devices
- More devices becoming networked to allow web access
- New category of devices to manage web traffic among similar servers: load balancers
- Use of operating systems like Windows 95, client-side, have evolved into Linux and Windows XP, which can be clients and servers



- **Distributed Computing Environment**

A distributed computing environment contains multiple nodes that are physically separate but linked together using the network. All the nodes in this system communicate with each other and handle processes in collaboration. Each of these nodes contains a small part of the distributed operating system software.

- **Cloud Computing Environment**

The computing is moved away from individual computer systems to a cloud of computers in cloud computing environment. The cloud users only see the service being provided and not the internal details of how the service is provided.

- **Cluster Computing Environment**

The clustered computing environment is similar to parallel computing environment as they both have multiple CPUs. However a major difference is that clustered systems are created by two or more individual computer systems merged together which then work parallel to each other.

Open-Source Operating Systems

"**open source**" refers to computer software or applications where the owners or copyright holders enable the users or third parties to use, see, and edit the product's source code.

- Apple's iOS, Microsoft's Windows, and Apple's Mac OS are closed operating systems.
- open-source operating systems are Linux, Open Solaris, Free RTOS, Open BDS, Free BSD, Minix, etc.

Best Open-Source Operating System

- Linux Kernel
- Linux Lite
- Linux mint
- Fedora
- React OS
- Solus
- Chrome OS